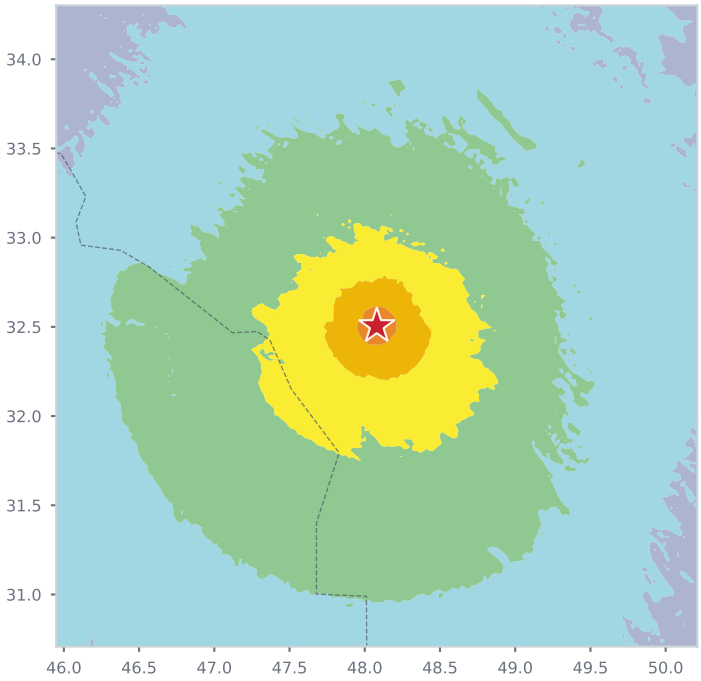
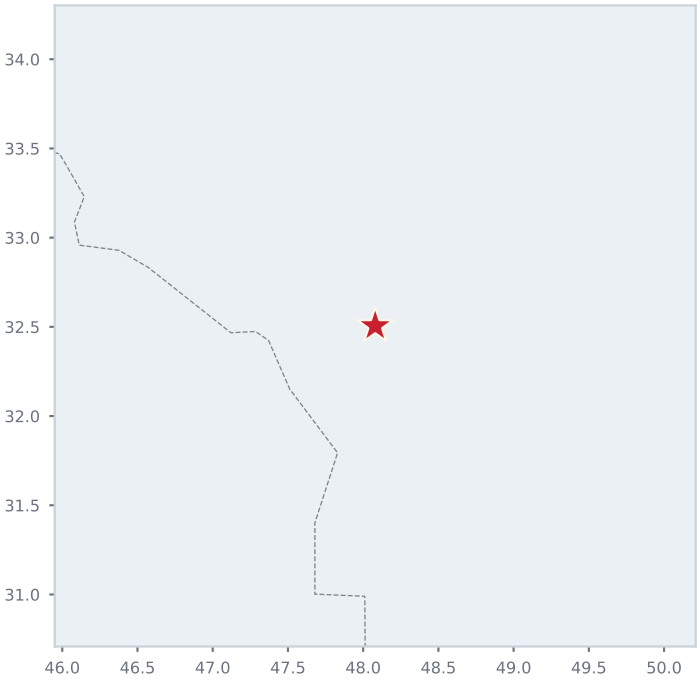
Shaking intensity (IMS-25)



Expected building damage grade



IMS-25 intensity

I II III IV V VI VII VIII IX X XI XII

Expected damage grade (IMS-25)

DG1 DG2 DG3 DG4 DG5

Damage alert

YELLOW Some buildings damaged, mostly repairable

Provisional **Transit occupancy**

Buildings heavily damaged (DG3 or worse)

190 to 3.7 thousand

Central estimate 780. The range is the 5th to 95th percentile.

Buildings in the shaken area

866 thousand

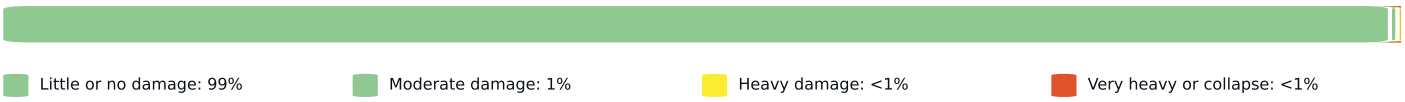
21% of them inferred from built-up area, not observed footprints.

People by shaking level

VI	23 thousand
V	1.3 million
IV	3.3 million

Building exposure modelled for: Iraq. Shaded areas without exposure data are shown for shaking only.

Share of buildings in the mapped area by expected damage



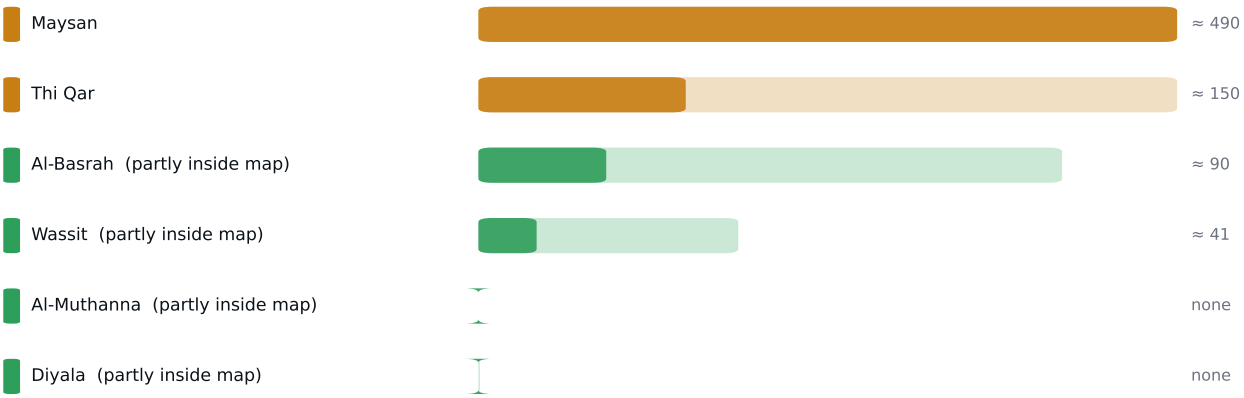
Little or no damage: 99% | Moderate damage: 1% | Heavy damage: <1% | Very heavy or collapse: <1%

Which buildings failed

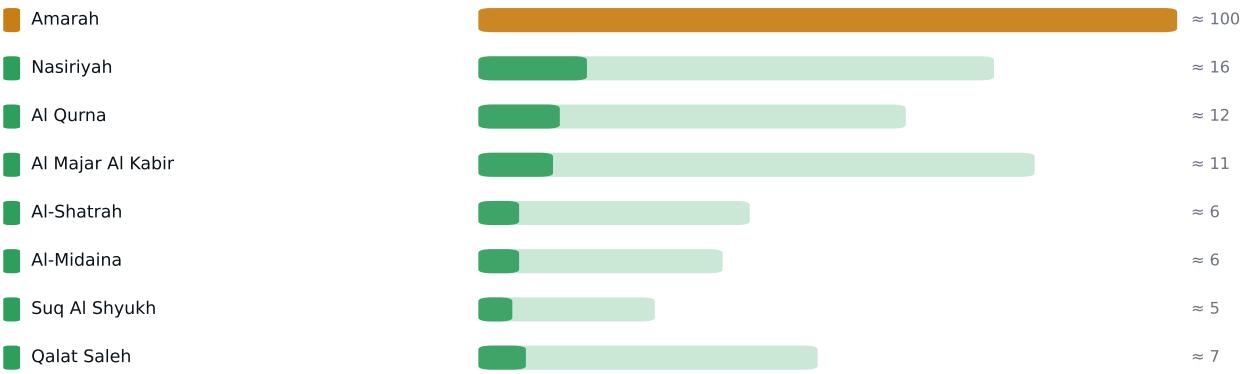
Share of each type's own buildings heavily damaged, worst first

Rubble stone, fieldstone	class A	**0.4%**	≈ 120 buildings
Adobe, earth brick	class A	**0.2%**	≈ 86 buildings
Manufactured stone units, timber floors	class B	**0.1%**	≈ 120 buildings
Simple stone	class B	**0.1%**	≈ 86 buildings
RC flat slab and others	class B	**0.1%**	≈ 1 buildings
Manufactured stone with RC floors	class C	**0.0%**	≈ 58 buildings
RC frame — no / low code (ERD-L)	class C	**0.0%**	≈ 12 buildings

Governorates most affected: heavily damaged buildings, central estimate and 90% range



Cities most affected



Every governorate inside the map

Area	Buildings	Heavy (P50)	90% range	% of its stock
Maysan	251 thousand	320	110 to 2.6 thousand	0.13%
Thi Qar	399 thousand	100	41 to 680	0.03%
Al-Basrah	152 thousand	59	22 to 410	0.04%
Wassit	63 thousand	27	11 to 180	0.04%
Diyala	840	none	none to 1	0%
Al-Muthanna	120	none	none to none	0%

How this report was made

Hazard engine Bumelerze SHAKEmap 26.2.0, ground-motion models CY14, ASB14, BSSA14, KALE15, intensity via Zaniniandhofer19 (IMS-25)

Observations 0 seismic stations, 0 felt-report cells (felt channel: none)

Damage scale IMS-25

Data sources

Built from USGS, EMSC and GEOFON earthquake parameters; building footprints from GFZ OpenBuildingMap (c) OpenStreetMap contributors, ODbL, and Microsoft, CDLA-Permissive-2.0; JRC Global Human Settlement Layer and WorldPop, CC BY 4.0; OCHA COD-AB Iraq, CC BY-IGO; GEM Global Exposure Model; IMS-25 vulnerability table. Full records, with licences and checksums: github.com/Peshawa-LH/bumelerze-engine

What this is, and what it is not

An automatic estimate of shaking and building damage from the earthquake's location and magnitude, adjusted with whatever station and felt-report data existed at the time.

Cite as: Hasan P. L. (2026) Bumelerze SHAKEmap and damage report, event bml19770001, version 2, generated 19 Oct 2026, 12:13 UTC.